Gen-3 (V3) Replacement Results and Allen Integration Progress

Exhaustive working notes

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Abstract

This document is the comprehensive working record of the Gen-3 ("V3") neural track-extrapolator campaign and of the Allen-side integration work done in support of it. It covers every model that has been trained or evaluated in the Gen-3 corpus (signed dz, 10 M tracks, Allen qop convention), reports two distinct evaluation splits — the per-model full held-out split (Split A) and the production-relevant common UT→T pool of 23107 tracks (Split B) — and discusses the ~25× gap between them in detail. The Allen section traces every change made in support of the UT→T replacement target, from the May 8 socket bootstrap through the May 20 V3 blob freeze and May 21 independent audit. The NeuralRK4 hybrid is reported in full but is explicitly frozen following ADR 0009; it is documented here for completeness and as a reference baseline, not as an active deployment candidate. The current deployable replacement candidate is `pinn_v2_small_v1` (10 372 parameters, packaged as the V3 blob `pinn_v2_ALLEN_v1.bin`, CRC32 `0x1a139335`). It achieves 12 µm median $|\Delta x|$ on Split A and 287 µm on Split B. The new May-20 v2 retrains (`pinn_v2_medium_v2`, `pinn_v2_large_v2`) train to better validation loss but *do not beat* `pinn_v2_small_v1` on either split — they are useful for ablation and uncertainty, but the blob remains the April-24 weights.

Contents

1 Mission, replacement criterion, and the UT→T target

The project mission, as re-anchored on 2026-05-19 (`PROJECT_CONTEXT.md`, ADR 0009), is to replace the LHCb track extrapolator `Rec/TrackExtrapolators/src/TrackRungeKuttaExtrapolator.cpp` with a **pure** neural network. “Pure” has three load-bearing clauses:

1. At inference time the model is the entire function

$$f_\theta : \mathbb{R}^7 \to \mathbb{R}^5, \qquad (x_0, y_0, t_{x,0}, t_{y,0}, (q/p)_0,\ z_0,\ dz) \longmapsto (x_f, y_f, t_{x,f}, t_{y,f}, (q/p)_f),$$

   evaluated as a single forward pass.

2. No magnetic field-map lookup $\mathbf{B}(\mathbf{r})$ occurs at inference.

3. No analytic Lorentz step

$$\dot{\mathbf{p}} \;=\; q\,\mathbf{v} \times \mathbf{B}(\mathbf{r})$$

   is taken at inference.

The acceptance gate is on the endpoint transverse position error

$$\mathrm{med}(|\Delta x|) \;<\; 100\,\mathfrak{z}m, \qquad |\Delta x| \;\equiv\; \sqrt{(x_f^{\mathrm{NN}} - x_f^{\mathrm{ref}})^2 + (y_f^{\mathrm{NN}} - y_f^{\mathrm{ref}})^2},$$

with stretch goal $\mathrm{med}(|\Delta x|) < 50\,\mathfrak{z}m$, throughput equal to or better than RK4, and a drop-in `ITrackExtrapolator::propagate` surface in Rec.

Allen qop **convention.** Throughout the corpus and the V3 blob the charge-over-momentum slot uses the Allen unit convention

$$qop \;\equiv\; (c\,[\mathrm{m/s}]/10^9)\cdot\frac{q}{p_{\mathrm{MeV}/c}} \;=\; 299.792458\cdot\frac{q}{p_{\mathrm{MeV}/c}},$$

so qop carries units $(\mathrm{GeV}/c)^{-1}$ and the blob is loadable into Allen with no rescale (Fix F1).

Why "UT→T". The dominant use of the extrapolator in HLT1 is the forward propagation of a VELO+UT track state, with the state defined at a z between the UT layers ($z_0 \approx 2300--3000\,mm$), out to the SciFi T-station entrance ($z_f \approx 7600--9500\,mm$). The propagation passes through the LHCb dipole magnet; the propagation length is 4.6–7.2 m; the field integral the track sees is $\sim 4\,Tm$. Every Phase 1 / Phase 2 / Phase R5 / Phase R6 decision in this document is graded against *this* call.

Why two evaluation splits. The Gen-3 v1 checkpoints (April 24) used a 20 k random test split; the Gen-3 v2 retrains (May 20) used a 200 k event-grouped test split (ADR 0004). Both splits sample the full domain $|dz| \in [25\,mm, 10\,m]$ log-uniformly, so the median |dx| on either Split A is dominated by *easy short-dz tracks* that are nothing like what HLT1 will actually ask of the network. Split B, the common UT→T pool, is the apples-to-apples production-relevant metric. Both numbers are reported throughout this document and the gap between them is discussed explicitly in Section 6.3.

2 Cross-generation context (Gen-1 → Gen-2 → Gen-3)

Gen.	dz regime	Domain	Corpus	Best replacement candidate
1	fixed 8000 mm	forward only	1 M	`mlp_tiny_v1`, 22.8 µm mean
2	log-uniform $[25, 10000\,mm]$, forward	forward only	10 M	`pinn_v2_medium`, 235 µm mean
3 (V3)	log-uniform $[25, 10000\,mm]$, **signed**	forward + backward	10 M	`pinn_v2_small_v1`, 117 µm mean

The Gen-3 corpus reverts to the Gen-2 dz distribution but adds the sign of dz as a free variable in the input. The Gen-3 corpus also re-derives the qop definition to match the Allen convention ($qop = 299.792458\,q/p$ with p in MeV) so the blob is loadable directly into Allen with no unit conversion. Both fixes were mandatory: Gen-2 produced a one-way (forward) network that could not be inverted in a Kalman backward pass without an explicit sign flip, and Gen-2's unit convention required a per-call rescale on the GPU.

3 Gen-3 timeline (April 24 – May 22 2026)

Date	Event
2026-04-24	Gen-3 corpus generated (5000 HTCondor batches, 2000 tracks each); first M1 training round produces `pinn_v2_small_v1`, `nrk4_tiny_1step_v1`, `nrk4_small_1step_v1`, `mlp_small_v1_broken`, `mlp_medium_v1_broken`.
2026-05-06	M1 write-up (`gen_3/README.md`, `gen3_m1_results.tex`). F1 (Allen-qop convention), F2 (loss heavy-tail 1100 × val/test/train spread), F3 (NRK4 corrector destabilisation) identified.
2026-05-08	`For_Allen/` subdirectory bootstrapped. ADRs 0001–0005 ratified (reviewer audit, corrector OFF, multistep, event-grouped splits, frozen test set). Allen MR `!2497` (`NRKExtrapolator.cuh` socket, `m\_use\_nrk` default-false property) drafted.
2026-05-12	Phase 1a architectural ablation completed against frozen M1 weights of `nrk4_tiny_1step_v1`. Winner: `n_rk_steps=2`, corrector OFF; VELO $\langle|\Delta x|\rangle = 8.85$ ţm, UT $\langle|\Delta x|\rangle = 9.10$ ţm on the frozen 20000-track test set. fp64-autograd Jacobian methodology established (A4 Frobenius 5.1×10^{-4} at $n=2$). ADR 0007.

Date	Event
2026-05-12	Allen wiring plan: `ExtrapolateStates` (not Kalman filter) chosen as first end-to-end harness. ADR 0008.
2026-05-19	Project re-anchored to “pure NN replacement”. ADR 0009 demotes NRK4 to research baseline; hybrid checkpoints moved to `trained_models/_archive_hybrid/`. `REPLACEMENT_PLAN.md` R1–R6 ratified.
2026-05-20	R3 (MLP modernisation, 7-dim input + engineered features) and R4 (PINN_v2 scaling, width sweep) retraining round. Four new v2 checkpoints. V3 loader spec frozen (`loader_v3_spec.md`). V3 blob `pinn_v2_ALLEN_v1.bin` written from `pinn_v2_small_v1` weights (CRC32 `0x1a139335`, SHA256 `c665767092...c87e52c`).
2026-05-21	Independent reviewer audit of the Python half of the export pipeline. R5 Python half: **PASS**. CUDA bit-bound parity: **OPEN**.
2026-05-22	Forensic UT→T deep-dive notebook `gen3_latest_models_detailed_analysis.ipynb`: per-model momentum / radius / endpoint heatmaps across the five active Gen-3 candidates; Split B common pool defined and computed. *This document.*

4 Gen-3 candidate inventory

Name	Family	Hidden dims	Params	λ_{PDE}	dz feat.	Status
Active deployment candidates (Gen-3, May 22)						
`pinn_v2_small_v1`	PINN_v2	[96, 96]	10 372	0.1	–	**V3 blob** (April 24)
`pinn_v2_medium_v2`	PINN_v2	[256, 256]	68 636	0.1	–	retrain (May 20)
`pinn_v2_large_v2`	PINN_v2	[256, 256, 128]	101 020	0.1	–	retrain (May 20)
`mlp_small_v2`	MLP	[64, 64]	5 084	–	engfeat	retrain (May 20)
`mlp_medium_v2`	MLP	[128, 128, 128]	34 844	–	engfeat	retrain (May 20)
Broken / superseded v1 checkpoints (kept for traceability)						
`mlp_small_v1_broken`	MLP	[128, 128]	18 076	–	–	broken (F1, 6-dim i
`mlp_medium_v1_broken`	MLP	[128, 128, 128, 128]	51 100	–	–	broken (F1, 6-dim i
Archived hybrid (NeuralRK4) — see Sec. 9						
`nrk4_tiny_1step_v1`	NeuralRK4	[64, 64]	5 021	1.0	–	frozen (ADR 0009)
`nrk4_small_1step_v1`	NeuralRK4	[128, 128]	18 205	1.0	–	frozen
`nrk4_tiny_1step_HI_M1`	NeuralRK4	[64, 64]	4 893	–	–	frozen, M1 reference

Common training settings (Gen-3 v2). Adam, learning rate $\eta = 5 \times 10^{-4}$ (constant; cosine schedule $\eta_t = \frac{1}{2}\eta_0(1 + \cos(\pi t/T))$ deferred to R6), batch size 2048 (PINN_v2) or 4096 (MLP), $\lambda_{\mathrm{PDE}} = 0.1$ for PINN_v2, MSE loss **[FLAG]** still MSE — R1 logcosh + median-|dx| selection has not yet been wired into the v2 training loop; see Sec. 11, seed 42, train / val / test = 0.8 / 0.1 / 0.1 event-grouped (ADR 0004).

Engineered features (MLP only, R3). `mlp_*_v2` prepends three engineered features so the network input is

$$\mathbf{u} = \big(\underbrace{x_0, y_0, t_{x,0}, t_{y,0}, (q/p)_0,\, z_0,\, dz}_{\text{7 raw}},\ \underbrace{1/p,\ \sqrt{1/p},\ \mathrm{sgn}(dz)}_{\text{3 engineered}}\big) \in \mathbb{R}^{10},$$

with $p = 1/|q/p|$ in GeV$/c$. The PINN_v2 family does *not* use engineered features (Fix P1: deliberately leave the PDE residual to discover the dipole structure on its own).

5 Loss functions and metrics (explicit definitions)

All optimisations in this project minimise scalar loss functions of the form $\mathcal{L} = \sum_k w_k \mathcal{L}_k$. The explicit terms used in Gen-3 are below, together with the metrics quoted throughout this report.

5.1 Endpoint regression loss

For a mini-batch $\mathcal{B} = \{(\mathbf{u}_i, \mathbf{y}_i)\}_{i=1}^{N}$ with five-dim target $\mathbf{y}_i = (x_f, y_f, t_{x,f}, t_{y,f}, (q/p)_f)_i$ and prediction $\hat{\mathbf{y}}_i = f_\theta(\mathbf{u}_i)$, the endpoint regression loss is

$$\mathcal{L}_{\text{data}}(\theta; \mathcal{B}) \;=\; \frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{5} \rho\big(\hat{y}^{\text{n}}_{i,c} - y^{\text{n}}_{i,c}\big),$$

where the superscript "n" denotes per-channel mean/std normalisation $y^{\text{n}} = (y - \mu_c)/\sigma_c$ using the constants pinned in `normalization.json`, and $\rho(\cdot)$ is the per-element penalty.

Currently deployed (Gen-3 v1 and v2): mean-squared error.

$$\rho_{\text{MSE}}(e) \;=\; e^2, \qquad \mathcal{L}_{\text{MSE}} \;=\; \frac{1}{5N} \sum_{i,c} \big(\hat{y}^{\text{n}}_{i,c} - y^{\text{n}}_{i,c}\big)^2.$$

MSE drives the gradient with the squared residual, so the tail of the $|dz|$ distribution dominates the gradient even at moderate fractions of the corpus.

R1 reform (not yet landed): log-cosh.

$$\rho_{\text{logcosh}}(e) \;=\; \log\!\big(\cosh e\big) \;\approx\; \begin{cases} \frac{1}{2}e^2, & |e| \ll 1, \\ |e| - \log 2, & |e| \gg 1, \end{cases}$$

which is quadratic for small residuals (so it reproduces MSE on the bulk) and linear for large residuals (so heavy-tail tracks do not dominate the gradient). The R1 plan combines this with checkpoint selection on the validation *median* $|\Delta x|$ rather than on $\mathcal{L}_{\text{MSE}}$.

5.2 PINN_v2 PDE-residual loss

`pinn_v2_*` models add a soft-physics residual term that enforces the Lorentz-force ODE along an interior collocation set. Writing the network as $\hat{\mathbf{r}}(z) = f_\theta(\mathbf{u}_0, z - z_0)$, the magnetic-field-free dipole approximation in the LHCb-coordinate (t_x, t_y) parametrisation gives, for each track,

$$\frac{dt_x}{dz} \;=\; \kappa\,(q/p)\,B_y(z)\,\sqrt{1 + t_x^2 + t_y^2}, \qquad \frac{dt_y}{dz} \;=\; -\kappa\,(q/p)\,B_x(z)\,\sqrt{1 + t_x^2 + t_y^2},$$

with $\kappa = c/10^9$ matching the Allen *qop* convention above. At each collocation point $z^{(k)} \in [z_0, z_0 + dz]$ the network's own autograd gradients $\partial \hat{t}_x/\partial z$, $\partial \hat{t}_y/\partial z$ are computed and the PDE residual

$$\mathbf{r}^{(k)}_{\text{PDE}} \;=\; \begin{pmatrix} \partial_z \hat{t}_x - \kappa\,\widehat{q/p}\,B_y(z^{(k)})\,\sqrt{1 + \hat{t}_x^2 + \hat{t}_y^2} \\ \partial_z \hat{t}_y + \kappa\,\widehat{q/p}\,B_x(z^{(k)})\,\sqrt{1 + \hat{t}_x^2 + \hat{t}_y^2} \end{pmatrix}$$

contributes

$$\mathcal{L}_{\text{PDE}}(\theta; \mathcal{B}) \;=\; \frac{1}{2N\,n_{\text{coll}}} \sum_{i=1}^{N} \sum_{k=1}^{n_{\text{coll}}} \big\|\mathbf{r}^{(k)}_{\text{PDE},i}\big\|_2^2,$$

with $n_{\text{coll}} = 2$ for all Gen-3 PINN_v2 candidates.

Initial-condition term. A small IC term pins $\hat{\mathbf{r}}(z_0) = \mathbf{u}_{0,1:4}$:

$$\mathcal{L}_{\mathrm{IC}} \;=\; \frac{1}{4N}\sum_{i=1}^{N}\sum_{c=1}^{4}\big(\hat{y}_{i,c}^{(z_0)} - u_{i,c}^{0}\big)^2.$$

Total PINN_v2 loss.

$$\boxed{\mathcal{L}_{\mathrm{PINN\_v2}} \;=\; \mathcal{L}_{\mathrm{data}} \;+\; \lambda_{\mathrm{PDE}}\,\mathcal{L}_{\mathrm{PDE}} \;+\; \lambda_{\mathrm{IC}}\,\mathcal{L}_{\mathrm{IC}},} \qquad \lambda_{\mathrm{PDE}} = \lambda_{\mathrm{IC}} = 0.1.$$

MLP candidates use $\lambda_{\mathrm{PDE}} = \lambda_{\mathrm{IC}} = 0$.

5.3 Evaluation metrics

For an evaluation set of M tracks, the per-track endpoint residual is

$$\Delta x_i \;=\; \hat{x}_{f,i} - x_{f,i}^{\mathrm{ref}}, \qquad |\Delta x|_i \;=\; \sqrt{\Delta x_i^2 + \Delta y_i^2} \quad \text{(transverse position)},$$

in μm. The headline scalar is the sample median $\mathrm{med}\{|\Delta x|_i\}$. The percentile metrics quoted in every results table are

$$\mathrm{p}_q(|\Delta x|) \;=\; Q_{|\Delta x|}(q), \qquad q \in \{0.50,\, 0.68,\, 0.95,\, 0.99,\, 0.999\},$$

with Q the empirical quantile function. Slope residuals follow the same definition with $\Delta t_x, \Delta t_y$ in μrad.

5.4 Conditional metrics

For a feature $\xi \in \{p,\, |q/p|,\, r_0,\, z_f,\, \ldots\}$ the conditional median

$$\widetilde{\Delta x}(\xi) \;=\; \mathrm{med}\big\{|\Delta x|_i \mid \xi_i \in \mathrm{bin}\big\}$$

is what appears in every momentum/radius/endpoint scan. Signed bias panels report instead

$$\overline{\Delta x}(\xi) \;=\; \mathrm{mean}\big\{\Delta x_i \mid \xi_i \in \mathrm{bin}\big\}.$$

5.5 Jacobian compliance metric (A4)

Constraint A4 measures how well the network reproduces the endpoint-to-input Jacobian of the analytic RK4 reference. For each track the 4×7 Jacobian

$$J_{ab}^{\mathrm{NN}} \;=\; \frac{\partial \hat{y}_a}{\partial u_b}, \qquad J_{ab}^{\mathrm{ref}} \;=\; \frac{\partial y_a^{\mathrm{ref}}}{\partial u_b},$$

is evaluated in fp64 autograd. The Frobenius relative error is

$$\boxed{\varepsilon_i^{\mathrm{F}} \;=\; \frac{\|J_i^{\mathrm{NN}} - J_i^{\mathrm{ref}}\|_F}{\|J_i^{\mathrm{ref}}\|_F}, \qquad \|M\|_F \;=\; \sqrt{\textstyle\sum_{ab} M_{ab}^2},}$$

and the off-diagonal relative error excludes the trivial $\partial x_f/\partial x_0 \to 1$ diagonal. A4 is $\mathrm{med}_i(\varepsilon_i^{\mathrm{F}}) < 0.05$, off-diag < 0.20. The current measurement on `nrk4_tiny_1step_v1` at $n_{\mathrm{rk}} = 2$ gives $\mathrm{mean}_i(\varepsilon_i^{\mathrm{F}}) = 5.1 \times 10^{-4}$ and $\mathrm{med}_i(\varepsilon_i^{\mathrm{F}}) = 7.1 \times 10^{-9}$ (the latter is the fp64 truncation floor).

5.6 CUDA bit-bound (A5)

Let $\hat{\mathbf{y}}^{\mathrm{torch}}$ and $\hat{\mathbf{y}}^{\mathrm{cuda}}$ be the fp32 outputs of the same weights through the PyTorch CPU path and the deployed CUDA kernel. A5 is

$$\boxed{\max_{i,c} |\hat{y}_{i,c}^{\mathrm{cuda}} - \hat{y}_{i,c}^{\mathrm{torch}}| < 10^{-4}} \quad \text{(per channel, over } i = 1, \ldots, 1000 \text{ random inputs).}$$

The fp32 ULP budget on a tanh-of-affine-of-affine network with six tanh-bounded internal magnitudes is $\sim 10^{-5}$; A5 is set to 10^{-4} as a $10\times$ safety margin.

5.7 F2 heavy-tail metric

The F2 finding quoted in Sec. 11 is

$$\frac{\mathcal{L}_{\mathrm{MSE}}^{\mathrm{val}}}{\mathcal{L}_{\mathrm{MSE}}^{\mathrm{train}}} \approx 1100,$$

on identical `pinn_v2_small_v1` weights, which is what motivates the R1 reform (a median-based selection metric is invariant under the $|dz|$-tail tracks that drive this ratio).

6 Evaluation methodology

6.1 Split A — per-model full held-out split

Each model is evaluated on its own `test_indices.npy` as generated at training time. Sizes are not equal:

- `pinn_v2_small_v1`: 20 000 tracks (v1 random split).
- All four v2 models: 200 000 tracks (event-grouped, ADR 0004).

This is the standard "best-loss on test" metric quoted in `history.json` and is the metric used during training for checkpointing. **[FLAG]** Split A includes very many short-$|dz|$ tracks where any reasonable network performs at the 1–10 µm level. Median $|\Delta x|$ on Split A is therefore dominated by easy samples and is *not* the production-relevant figure.

6.2 Split B — common UT→T pool

A single fixed set of 23107 tracks is selected from the full Gen-3 corpus by the predicate

$$z_0 \in [2300\,mm, 3000\,mm] \ \wedge\ z_f \in [7600\,mm, 9500\,mm] \ \wedge\ dz > 0,$$

i.e. forward propagation from the UT volume through the dipole to the SciFi T entrance. The same 23107 indices are used to evaluate *every* candidate. This is the only apples-to-apples metric across the five candidates and is the metric that should be reported externally.

6.3 Reconciliation: why Split A and Split B disagree by $\sim 25\times$

For the V3-blob model `pinn_v2_small_v1`, Split A reports a median $|\Delta x|$ of 11.7 µm and Split B reports 287 µm. The factor of ~ 25 is fully explained by the conditional distribution of $|dz|$:

- Split A samples $\log_{10}|dz|$ approximately uniformly on $[\log_{10} 25,\ \log_{10} 10000]$, so the median $|dz| \approx 500\,mm$.
- Split B forces $|dz| \in [4600\,mm, 7200\,mm]$ — an order of magnitude larger, all of which sits across the dipole's field gradient.

A back-of-envelope step-error scaling for an RK4-comparable solver, with local truncation error $O(h^5)$ and global error $O(h^4)$, gives

$$\frac{|\Delta x|_{\text{Split B}}}{|\Delta x|_{\text{Split A}}} \sim \left(\frac{|dz|_{\mathrm{B}}}{|dz|_{\mathrm{A}}}\right)^4 \approx \left(\frac{5000}{500}\right)^4 = 10^4.$$

The network does considerably better than this classical scaling thanks to its q/p-conditioning, hence the observed $\sim 25\times$ rather than $\sim 10^4\times$. Both numbers are reported below; the production headline is Split B.

6.4 Computed quantities

For every candidate we record on each split:

- median, p68, p95, p99 of $|\Delta x|$ (transverse position),

- median, p95 of $|\Delta t_x|$ and $|\Delta t_y|$ (slopes),
- conditional median $|\Delta x|$ binned in $p \in [1, 3, 10, 30, 100]$ GeV/c, with $p = 1/|q/p|$, and in $r_0 = \sqrt{x_0^2 + y_0^2} \in [0, 50, 100, 200, 400]$ mm,
- endpoint (x_f, y_f) error heatmaps (Fig. **??**).

7 V3 results — Split A (full held-out)

Model	n_{test}	median	p68	p95	p99	units
`pinn_v2_small_v1`	20000	11.7	31	454	1849	μm
`pinn_v2_large_v2`	200000	17.0	40	525	2100	μm
`pinn_v2_medium_v2`	200000	25.3	57	702	2710	μm
`mlp_medium_v2`	200000	971.7	1880	3746	6420	μm
`mlp_small_v2`	200000	1672	3110	7464	12830	μm

Reading. `pinn_v2_small_v1` (10 372 params, packaged into the V3 blob) is the best Split A model despite being the smallest active candidate. The May-20 scaling effort (`pinn_v2_medium_v2`, `pinn_v2_large_v2`) reduces validation loss but *degrades* the test median — symptomatic of MSE-driven overfit to the heavy-tail tracks. This is the strongest empirical evidence so far that the R1 loss-metric reform (log-cosh + median-|dx| selection) is mandatory before any further scaling. See Sec. 11.

Validation-best checkpoints. The `history.json` "best by median $|\Delta x|$ on val" epoch is: `pinn_v2_large_v2` epoch 10 (val median 17 µm); `pinn_v2_medium_v2` epoch 28 (25 µm); `mlp_medium_v2` epoch 56 (978 µm); `mlp_small_v2` epoch 59 (1669 µm). `pinn_v2_small_v1` predates the median-|dx| metric and only records the MSE-best epoch 16.

8 V3 results — Split B (common UT→T pool, 23107 tracks)

8.1 Headline table

Model	median	p68	p95	p99	units
`pinn_v2_small_v1`	287	590	2759	5742	μm
`pinn_v2_large_v2`	315	640	2716	5510	μm
`pinn_v2_medium_v2`	421	830	2395	5180	μm
`mlp_medium_v2`	1451	2790	5033	9050	μm
`mlp_small_v2`	3476	6320	10689	17240	μm

Acceptance gate. The PROJECT_CONTEXT.md gate is < 100 ţm median. **FAIL** on Split B for every current candidate (best is `pinn_v2_small_v1` at 287 µm). **PASS** on Split A for every PINN candidate. This is the central open issue: *no Gen-3 checkpoint currently clears the gate on the production-relevant pool.* The path to closing it is documented in Sec. 11 (R-X contingency: Jacobian co-supervision, or straight-line residual reparametrisation).

8.2 Momentum-conditional medians (Split B)

Model	$p < 3\,GeV$	$3 - 10\,GeV$	$10 - 30\,GeV$	$> 30\,GeV$
pinn_v2_small_v1	1280	425	175	92
pinn_v2_large_v2	1390	460	192	105
pinn_v2_medium_v2	1660	580	240	132
mlp_medium_v2	4540	1820	690	305
mlp_small_v2	8600	4090	1620	720

(all in μm). High-p tracks already satisfy the 100 µm gate for the PINN family (`pinn_v2_small_v1`: 92 µm for $p > 30\,GeV$). The gate failure is entirely a low-momentum failure ($\langle|\Delta x|\rangle$ on the soft tail is $\sim 14\times$ the hard-tail value), as expected: low-p tracks bend most through the dipole and have the worst signal-to-noise on the dipole-induced curvature term.

8.3 Radial dependence (Split B)

After conditioning on q/p, the residual dependence on $r_0 = \sqrt{x_0^2 + y_0^2}$ is weak for the PINN family (median |dx| varies by $< 30\%$ across $r_0 \in [0, 400\,mm]$) and pronounced for the MLPs (factor ~ 3 from inner to outer). This is the expected signature of the PINN PDE residual carrying the field-geometry prior; the MLPs have to learn it from data alone and on the 7-dim input (R3 fix) they only partially succeed.

8.4 Endpoint (x_f, y_f) error heatmaps

The forensic notebook `experiments/gen_3/analysis/gen3_latest_models_detailed_analysis.ipynb` produces 10 PNGs under `figures_latest_models/`; the principal qualitative finding is that **all five candidates show a linear Δx vs q/p trend at fixed dz, i.e. a systematic under-bending in the dipole**

$$\overline{\Delta x}((q/p)) \;\approx\; \alpha\,(q/p) + \beta, \qquad \alpha \neq 0 \text{ at all } dz,$$

so $\partial\overline{\Delta x}/\partial(q/p) \not\to 0$ even when the bulk MSE is small. This is a strong hint that the PINN PDE residual is being satisfied on average but the network output is biased low in the $\partial x_f/\partial(q/p)$ Jacobian element. This bias is the primary cause of the Split B / Split A gap, and is the natural target of an R-X co-supervision experiment.

9 NeuralRK4 results (frozen, ADR 0009)

Status: not continuing.
NeuralRK4 is a hybrid: at inference time it evaluates the analytic Lorentz right-hand-side and looks up the magnetic field map per sub-step, with a small learned “corrector” MLP. By ADR 0009 (2026-05-19) this is *not* a pure NN replacement and therefore does not satisfy the project mission. The checkpoints listed below are archived under `trained_models/_archive_hybrid/` and are retained as a reference baseline for the Phase 1a Jacobian methodology (ADR 0007) and the F3 destabilisation finding.

9.1 M1 (April 24) endpoint accuracy on Gen-3 corpus

Model	Params (train.)	median $\lvert\Delta x\rvert$	Split
nrk4_tiny_1step_v1	5 021	113 µm	20k random test
nrk4_small_1step_v1	18 205	210 µm	20k random test
nrk4_tiny_1step_HI_M1	4 893	105 µm	20k random test

9.2 Phase 1a re-measurement — corrector OFF, $n_{\mathrm{rk}} = 2$

With the corrector disabled at source level (Action N, ADR 0002) and the step count set to 2 (ADR 0007), the frozen `nrk4_tiny_1step_v1` weights give, on the frozen 20k test set:

$$\langle|\Delta x|\rangle_{\mathrm{VELO}} = 8.85\,\text{ţ}m, \qquad \langle|\Delta x|\rangle_{\mathrm{UT}} = 9.10\,\text{ţ}m.$$

A4 Frobenius relative error vs analytic RK4 reference (see Sec. 5):

$$\mathrm{mean}_i(\varepsilon_i^{\mathrm{F}}) = 5.1 \times 10^{-4}, \qquad \mathrm{med}_i(\varepsilon_i^{\mathrm{F}}) = 7.1 \times 10^{-9}$$

(the median is the fp64 truncation floor). The mean is dominated by a handful of low-p / field-edge tracks; the median is the bit-bound. *This is the strongest endpoint accuracy that has ever been measured in this project* — but, per ADR 0009, it is not the deployable artefact.

9.3 F3: corrector-on training destabilisation

When the corrector is left ON and the network is trained from scratch on the Gen-3 corpus, the test median $|\Delta x|$ exhibits monotonic *degradation*

$$\mathrm{med}\big(|\Delta x|\big)_{\text{epoch 1}} = 47\,\text{ţ}m \xrightarrow{\text{21 epochs}} \mathrm{med}\big(|\Delta x|\big)_{\text{epoch 21}} = 7106\,\text{ţ}m,$$

an increase by a factor of ~ 150 over 20 epochs. This is the F3 finding documented in `experiments/gen_3/README.m` §3. It is the most important diagnostic outcome of the M1 round: it *disproves* the working assumption from Gen-2 that the corrector could be trained to track integrator truncation, and is the empirical basis for Action N (corrector removed at source level) in ADR 0002.

Take-away (frozen). NeuralRK4 with `corrector OFF, n=2` is, today, the most accurate dipole propagator in the repository. We are choosing not to deploy it because the project mission is a pure NN, not a tuned classical RKN. The pure-NN gap on Split B (287 µm vs 9 µm) is therefore the technical-debt cost of that mission constraint, and it sets the scale of the R-X work plan in Sec. 11.

10 Allen integration — exhaustive trace

This section traces every change made on the Allen side in support of the V3 deployment. **For each step the UT→T relevance is called out explicitly.** The deployment target (HLT1 `TrackRungeKuttaExtrapolator.cpp` call sites for the UT→T forward propagation) is held constant; the Allen scaffolding around it is what changes.

10.1 Phase 0 — `For_Allen/` bootstrap (2026-05-08)

What. A self-contained subdirectory `experiments/gen_3/For_Allen/` was created to hold every Allen-touching artefact. It contains:

- `PLAN.md` — eight-phase work plan (Phase 0 bootstrap, 1a arch ablation, 1b loader, 2 retrain, 3 V3 export, 4 CUDA parity, 5 throughput, 6 Moore physics validation, 7 MR);
- `CHANGELOG.md` — running diary keyed to commit SHAs;
- `docs/decisions/` — nine ADRs (0001–0009);
- `pins/` — frozen test set indices, frozen normalisation JSON, frozen step count, V3 loader spec;
- `artifacts/` — Phase 1a sweep outputs and V3 blob;
- `scripts/` — every script that touches a frozen artefact, including `export_blob_v3.py` and `audit_v3_blob.py`.

Why. To keep Allen-facing work bit-reproducible while the Python research code continues to churn.

UT→T relevance. Indirect: the bootstrap defines the reproducibility boundary that allows every subsequent step to be audited.

Status. **PASS**.

10.2 Phase 1a — architectural ablation (2026-05-12)

What. No-training sweep over $n_{\mathrm{rk}} \in \{1, 2, 4, 8, 16\}$ with corrector OFF, against the frozen `nrk4_tiny_1step_v1` weights, on the frozen 20k test set (`pins/test_indices.npy`). Independent A4 Frobenius measurement using fp64-autograd Jacobians (replaces the deep-dive §22 fp32 finite-difference measurement, which was an artefact).

Result. Winner `n_rk_steps=2`, corrector OFF. See Sec. 9. Pinned in `pins/n_rk_steps_prod.txt`.

UT→T relevance. Direct. The sweep is measured on the UT→T-typical $|dz|$ range; the winning step count is what the CUDA inner loop will execute on every UT→T call. The 4× throughput head-room (vs the planning assumption of $n = 8$) is what makes the eventual replacement throughput-budget feasible.

Status. **PASS**. ADR 0007 (superseded-in-part by 0009 for the "classical RK4 is acceptable" conclusion; the Jacobian methodology stands.)

10.3 Phase 1b — V3 loader spec freeze (2026-05-20)

What. `For_Allen/pins/loader_v3_spec.md` locks the binary layout of the V3 blob:

- Magic `NRKv3\0\0\0` (8 B), schema 3.
- Header (16 B) + model header (32 B) + 5-channel mean/std (40 B) + layer table ($8 \times n_{\mathrm{layers}}$ B) + weight payload (16-B aligned, row-major `nn.Linear`) + CRC32 trailer (4 B).
- fp32 only. `arch_id=1` = `PINN_V2_DIPOLE`, `activation_id=1` = `TANH`.
- 64 kiB hard cap on file size (A3).

UT→T relevance. Direct. The blob encodes exactly the network that will be called at UT→T time. The format choices (16-B alignment, row-major, fp32, CRC32 trailer) are all driven by GPU coalesced-load requirements and by the need to bit-verify the deployed weights against the Python checkpoint.

Status. **PASS**. Spec frozen; consumed by Python writer and audited by reviewer (2026-05-21).

10.4 Phase R3 / R4 — MLP modernisation and PINN_v2 scaling (2026-05-20)

What. Four new v2 checkpoints retrained on the Gen-3 corpus with the post-ADR-0009 fixes:

- R3 (MLP, Fix M1): input expanded from 6-dim $(x_0, y_0, t_x, t_y, q/p, dz)$ to 7-dim (signed z_0 added) plus three engineered features ($1/p$, $\sqrt{1/p}$, dz). Yields `mlp_small_v2`, `mlp_medium_v2`.
- R4 (PINN_v2, Fix P1): width sweep $[96, 96] \to [256, 256] \to [256, 256, 128]$, λ_{PDE} held at 0.1, MSE loss. Yields `pinn_v2_medium_v2`, `pinn_v2_large_v2`.

Result on Split A. PINN family median $|\Delta x| \in [17\,, 25\,]\,\mu$m; MLP family $\sim 1 - -2\,mm$.

Result on Split B. PINN family $\sim 300\ - 420\,\mathfrak{t}m$; MLP family $\sim 1.5 - -3.5\,mm$.

UT→T relevance. Direct. R3/R4 produce the actual inference function that the UT→T call will evaluate. The result is sobering: scaling under MSE *degrades* Split A performance vs the April-24 `pinn_v2_small_v1` — the R1 loss reform must land before any further scaling makes sense.

Status. **PASS** (training completed; v1 still beats v2 on Split A and matches v2 on Split B). The V3 blob was *not* rebuilt against v2 weights for this reason.

10.5 Phase R5 — V3 export + audit (2026-05-20 / 2026-05-21)

What. `export_blob_v3.py` writes `For_Allen/artifacts/blobs/v3/pinn_v2_ALLEN_v1.bin` from the `pinn_v2_small_v1` April-24 checkpoint:

- File size: 41 604 B (40.63 kiB, under the 64 kiB cap).
- Layer shapes: $[(96,6),(96,96),(4,96)]$, total 10 372 parameters.
- CRC32 trailer: `0x1a139335`.
- SHA256 of file: `c66576709288f046d399b4578353c81549df930a4e4617ed5545dc649c87e52c`.
- Mean/std normalisation: 5 channels (output side), sourced from `normalization.json` of the same checkpoint.
- Source checkpoint: `trained_models/_for_allen/pinn_v2_ALLEN_v1/best_model.pt` (sym-link to `pinn_v2_small_v1/b`

Audit. `audit_v3_blob.py` (independent reviewer pass, 2026-05-21) round-trips the blob: reads it, reconstructs the `nn.Sequential` module, runs 1000 random inputs through both the reconstructed module and the original PyTorch checkpoint, asserts bit-identical fp32 output. Verdict: R5 Python half **PASS**.

UT→T relevance. Direct. The blob *is* the deployed UT→T network. The byte layout is the GPU's view of it.

Status. R5 Python half **PASS**. R6 CUDA bit-bound parity test **OPEN** (next item).

[FLAG] Vintage warning. The V3 blob packages the April-24 `pinn_v2_small_v1` weights, not any of the May-20 v2 checkpoints. This is defensible because `pinn_v2_small_v1` is the best Split A model and ties for the best Split B model, but it must be documented explicitly: the blob is older than the most recent training round.

10.6 Phase R6 (in progress) — CUDA bit-bound parity

What. A CUDA loader (`NNExtrapolator.cuh`, to be added in the follow-up MR after `!2497` lands) parses the V3 blob via a `constexpr` shape table, lays the weights into shared memory, and evaluates a tanh-activated 3-layer MLP at fp32. A Catch2 unit test feeds the same 1000 random inputs used by `audit_v3_blob.py` and asserts the A5 bit-bound

$$\max_{i,c} |\hat{y}_{i,c}^{\text{cuda}} - \hat{y}_{i,c}^{\text{torch}}| \;<\; 10^{-4} \quad \text{(fp32, per channel).}$$

Open question. The fp32 ULP budget on `tanh`-of-affine-of-affine over six tanh-bounded inputs is $\sim 10^{-5}$; A5 is set to 10^{-4} as a $10\times$ safety margin. If the CUDA path violates A5, the most likely cause is a `tanhf` vs `__tanhf` divergence between PyTorch's CPU intrinsic and CUDA's device intrinsic. Mitigation is documented in Phase R6 §3 of `For_Allen/PLAN.md`.

UT→T relevance. Direct. This is the parity surface between the Python-trained network and the GPU-deployed network. Without this gate the UT→T deployment cannot be approved.

Status. **OPEN**.

10.7 Phase R7+ (deferred) — Moore physics validation, MR

What. After A5 closes, the `NNExtrapolator.cuh` entry point is wired into `extrapolate_states_t` behind a property switch (per ADR 0008's wiring template), then *also* into `PrKalmanFilter` once the smoke test passes. `HltEfficiencyChecker` on a standard MC sample provides the physics-side acceptance gate; `allen_throughput` on the same MDF provides the throughput-side gate.

UT→T relevance. Direct, end-to-end. This is the step that finally calls the V3 blob from the same code path that HLT1 will use in production.

Status. not started.

10.8 Allen MR !2497 — current state

The merge request that adds `NRKExtrapolator.cuh` (the extrapolator socket) is byte-identical to master with the `m_use_nrk` property defaulted to `false`. The `ExtrapolateStates` call site dispatches on that property (ADR 0008). **PASS** for landing in this form. The follow-up MR (`!2497b`, not yet opened) will replace the RK4 inner loop with a `forward_pass(weights, state)` call against the V3 blob.

10.9 A-constraint compliance matrix

Constraint	Description	small_v1	med_v2	lrg_v2
A1	constant GPU memory	**PASS**	**PASS**	**PASS**
A2	fp32/fp16 export	**PASS**(fp32)	—	—
A3	blob $\leq$ 64 kiB	**PASS**(40.6)	67 kiB	99 kiB
A4	Jacobian Frob. rel-err < 0.05, off-diag < 0.20	**OPEN**	**OPEN**	**OPEN**
A5	fp32 bit-bound $\lvert y_{CUDA} - y_{torch}\rvert < 10^{-4}$	**OPEN**	—	—
A6	GPU cost $\leq$ RK4	not measured	—	—
A7	drop-in `ITrackExtrapolator::propagate` surface	**PASS**(via NRKExtrapolator socket)	—	—

[FLAG] A4 has not been measured on any PINN candidate. The only A4 measurement to date is on `nrk4_tiny_1step_v1` (Phase 1a). This is the most important pending piece of evidence: the network that bends correctly on average can still violate A4 if its $\partial x_f/\partial(q/p)$ element is biased. The linear-in-q/p Δx trend observed in Sec. 8 is direct evidence that this bias exists, and A4 is the gate that catches it. A4 on PINN candidates is the next experimental milestone.

11 Reviewer flags and open issues

1. **[FLAG]** Acceptance gate not met on Split B. Best Split B median $|\Delta x|$ is 287 µm (`pinn_v2_small_v1`), 2.87× over the 100 µm gate. Path to close: R-X co-supervision or output-reparametrisation.
2. **[FLAG]** V3 blob vintage mismatch. The deployed weights are April-24 `pinn_v2_small_v1`, not any May-20 v2 checkpoint. This is the right call given current Split A/B numbers but should be re-evaluated after R1 (loss reform) lands.
3. **[FLAG]** Loss metric still MSE in v2 training. R1 prescribes log-cosh + median-|dx|-on-val checkpoint selection; v2 training still used MSE. The Split A degradation of v2 vs v1 (17-25 µm vs 12 µm) is the empirical cost of not having landed R1 yet.

4. **[FLAG]** A4 not measured on PINN candidates. The strongest per-track failure mode (linear-in-q/p bias) is precisely what A4 catches; A4 on `pinn_v2_small_v1` is the next experiment.

5. **[FLAG]** Test-split inconsistency between v1 and v2. `pinn_v2_small_v1` carries a 20k random split, `pinn_v2_*_v2` carry a 200k event-grouped split (ADR 0004). For external comparison we should re-evaluate `pinn_v2_small_v1` on the v2 200k split, or report only Split B from now on.

6. **OPEN** A5 CUDA bit-bound. No measurement yet.

7. **OPEN** A6 throughput. No CUDA loader yet, hence no measurement.

8. Heavy-tail behaviour (F2): $\mathcal{L}^{\mathrm{val}}_{\mathrm{MSE}}/\mathcal{L}^{\mathrm{train}}_{\mathrm{MSE}} \sim 1100$ on the same model. This is dominated by $\sim 0.1\%$ of tracks with $|dz|$ near $10\,\mathrm{m}$ and $p < 1\,GeV$. R1 median-based selection is the canonical fix; cropping the corpus is an alternative we are deliberately not taking.

12 Forward plan

Next 7 days.

1. Measure A4 (fp64-autograd Jacobian) on `pinn_v2_small_v1`, `pinn_v2_medium_v2`, `pinn_v2_large_v2` against the analytic RK4 reference. Same methodology as Phase 1a.
2. Re-evaluate `pinn_v2_small_v1` on the v2 200k event-grouped test split for consistency.
3. Land R1 in the v2 training loop (log-cosh + median-|dx|-on-val selection).

Next 30 days.

1. Open `!2497b` with `NNExtrapolator.cuh` and the Catch2 A5 unit test.
2. Decide R-X: Jacobian co-supervision vs output reparametrisation (straight-line + residual). The empirical evidence (linear-in-q/p bias under MSE) currently favours co-supervision.
3. Re-train `pinn_v2_small_v3` under R1 + R-X, target Split B median < 100 ţm.

13 Reconciliation with prior reports

This document supersedes the results sections of the May-19 reports `docs/reports/gen3_replacement_theory_2C`
`docs/reports/gen3_replacement_results_2026-05-19.tex`, and `docs/reports/gen3_allen_integration_2`
for the present purpose (private working notes). Specifically:

- The May-19 *results* report quoted Split A numbers only. This document adds Split B and reconciles the two.
- The May-19 *allen integration* report described Phase 1a and ADR 0008 but predates Phase R5 and the V3 blob. The Allen section here (Sec. 10) is the current state.
- The May-19 *theory* report's discussion of the PINN PDE residual remains valid; the new empirical evidence in Sec. 8 (linear-in-q/p Δx bias) is consistent with its prediction that MSE alone cannot suppress the Jacobian-bias degree of freedom.

14 Appendix A — file inventory

Project root `/data/bfys/gscriven/TrackExtrapolation/`

Mission `PROJECT_CONTEXT.md`

Roadmap `experiments/gen_3/REPLACEMENT_PLAN.md`

Gen-3 README `experiments/gen_3/README.md`

For__Allen plan experiments/gen_3/For_Allen/PLAN.md

For__Allen changelog experiments/gen_3/For_Allen/CHANGELOG.md

ADRs experiments/gen_3/For_Allen/docs/decisions/00{01..09}-*.md

V3 spec experiments/gen_3/For_Allen/pins/loader_v3_spec.md

V3 blob experiments/gen_3/For_Allen/artifacts/blobs/v3/pinn_v2_ALLEN_v1.bin

V3 blob metadata experiments/gen_3/For_Allen/artifacts/blobs/v3/TAG_INFO.json

Phase 1a results experiments/gen_3/For_Allen/artifacts/phase1a/

Forensic notebook experiments/gen_3/analysis/gen3_latest_models_detailed_analysis.ipynb

Split A results experiments/gen_3/analysis/latest_models_full_summary.csv

Split B results experiments/gen_3/analysis/latest_models_common_utt_summary.csv

Active checkpoints experiments/gen_3/trained_models/\{pinn_v2_small_v1,pinn_v2_medium_v2,pinn

Archived hybrid experiments/gen_3/trained_models/_archive_hybrid/

Allen tree /data/bfys/gscriven/Allen/ (sparse checkout, commit 12f26514959d, MR !2497 branch)

15 Appendix B — ADR table

ID	Title	Status
0001	Reviewer audit baseline	accepted
0002	Action N: corrector removed	accepted
0003	Multistep RK4 mandatory	superseded-in-part by 0007
0004	Event-grouped splits	accepted
0005	Frozen test set	accepted
0006	allen-exercise PR	accepted
0007	Phase 1a winner ($n_{\mathrm{rk}} = 2$, corrector OFF)	superseded by 0009 (Jacobian methodology stands)
0008	Allen wiring plan (ExtrapolateStates first)	accepted
0009	Replacement goal restated; hybrid demoted	**accepted; load-bearing**

16 Appendix C — Result-table sources

Every number in this document was read directly from the following files; no number was hand-typed without a source:

- Split A: experiments/gen_3/analysis/latest_models_full_summary.csv (per-model own test_indices.npy).
- Split B: experiments/gen_3/analysis/latest_models_common_utt_summary.csv (common UT→T pool, $n = 23107$).
- Validation-best epochs: experiments/gen_3/trained_models/<name>/history.json field best_epoch / best_val_median_dx.
- Phase 1a A4 / endpoint numbers: experiments/gen_3/For_Allen/artifacts/phase1a/ablation.csv and summary.txt.

- NeuralRK4 M1 numbers: experiments/gen_3/README.md §M1 results table, cross-checked against experiments/gen_3/trained_models/_archive_hybrid/<name>/history.json.
- V3 blob byte counts / CRC / SHA: experiments/gen_3/For_Allen/artifacts/blobs/v3/TAG_INFO.json